DELIVERABLE G
Economics Report
Material Costs (per unit)

1. Telescopic Carbon Rods
    - Cost type: Variable, Direct, Material
    - Prototyping: $30
    - High-Volume Manufacturing: $15
2. Magnetic Tethers
    - Cost type: Variable, Direct, Material
    - Prototyping: $5
    - High-Volume Manufacturing: $1-2
3. Button Control System (electronics, button, wiring)
    - Cost type: Variable, Direct, Material
    - Prototyping: $40
    - High-Volume Manufacturing: $20-30
4. Air Pump Mechanism (pump, value, hose)
    - Cost type: Variable, Direct Material
    - Prototyping: $50
    - High-Volume Manufacturing: $15-25
5. Battery
    - Cost type: Variable, Direct Material
    - Prototyping: $5-10
    - High-Volume Manufacturing: $5
6. Belt
    - Cost type: Variable, Direct Material
    - Prototyping: $5-10
    - High-Volume Manufacturing: $2-5

Total Estimated Material Cost

- Prototype: $160
- High-Volume Manufacturing: $85

Labor Costs

1. Assembly Labor
    - Cost type: Variable, Direct, Labor
    - Prototype:$40-60 per hour
        - 1-2 hours per unit for assembly
    - High-Volume Manufacturing: $20-30
2. Quality Control
    - Cost type: Variable, Direct, Labor
    - Prototyping: $15-25

        - High-Volume Manufacturing: $10-15
3. R&D
    - Cost type: Fixed, Indirect, Labor
    - Prototyping: $5000 (one-time cost)
    - High-Volume Manufacturing: $1000

Total Estimated Labor Cost

- Prototyping: $150
- High-volume Manufacturing: $30-40

Overhead Costs

1. Prototyping equipment and materials
    - Cost type: Fixed, Indirect, Overhead
    - Prototyping: $500-1000 (is custom tooling is needed)
    - High-Volume Manufacturing: in initial setup
2. Manufacturing equipment (high volume)
    - Cost type: Fixed, Indirect, Overhead
    - Prototyping: $0
    - High-Volume Manufacturing: $2000-5000 (molds, tools, etc)
3. Utilities and Facility (per month)
    - Cost type: Variable (utilities) / Fixed (facility), Indirect, Overhead
    - Prototyping: $50-100
    - High-Volume Manufacturing: $500-1000
4. Packaging and shipping
    - Cost type: Variable, Indirect, Overhead
    - Prototyping: $5-10
    - High-Volume Manufacturing: $2-5

Total Estimated Overhead Cost (per unit)

- Prototyping: $100-200
- High-Volume Manufacturing: $10-20

Additional Costs

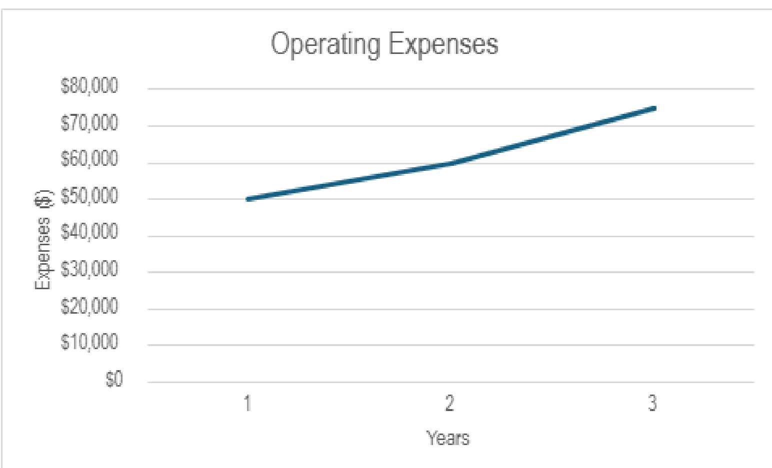
1. Marketing and distribution
    - Cost type: Fixed, Indirect, Labor
    - Prototyping: $50-100
    - High-Volume Manufacturing: $500-1000

3 Year Income Statement

Unit Price:$300

Year	**1**	**2**	**3**
Units Sold	200	500	1000
Sales Revenue	$60000	$150000	$300000
Costs of Goods Sold	$30000	$75000	$150000
Gross Profit	$30000	$75000	$150000
Operating Expenses	$50000	$60000	$75000
Operating Income	-$20000	$15000	$75000

Cash Flow Diagrams



NPV Analysis

Year	**1**	**2**	**3**
Fixed cost	$5500	$3000	$3000
Units sold for break-even	19	10	10

Assumptions for economic report

1. Market demand
    - Assumption: individuals with limited mobility, approx 50000 in Canada
    - Justification: based on stats and growing demand for assistive devices
2. Market Share
    - Assumption: capturing less than 1% of market in first 3 year
    - Justification: being conservative for early market incorporation with opportunity to grow
3. Unit Price
    - Assumption: $300 for first 3 years
    - Justification: competitive for small assistive device but still affordable compared to employing care worker
4. Sale Growth
    - Assumption: 250% annual sale growth for 3 years
    - Justification: growth demand as increase of product awareness
5. Fixed Costs
    - Assumption: stay $50000 annually for 3 years
    - Justification: for overhead and initial R&D

Intellectual Property Report

Patent: "Electromagnetic Gripping Apparatus for Supporting Objects" – US9732554B2

- Link: US9732554B2 on Google Patents
- Summary: This patent covers an electromagnetic gripping device that enables secure, temporary holding of objects, activated by an electric current. The technology includes specific configurations for generating a strong magnetic grip and mechanisms for releasing objects. Although not specifically designed as an assistive dressing device, this

patent is relevant to our project since it involves using electromagnets to grip and release, similar to the function we aim to achieve with our assistive pant device.

- Legal Constraints: Given the similarities in using electromagnets, we must ensure that our device does not replicate any unique mechanisms or configurations described in this patent. To avoid infringement, we explored alternative designs in magnetic force control, such as adjusting intensity or modifying the structure. Differentiating the functionality of the electromagnetic elements would help ensure our design remains compliant with IP laws. We would avoid issues for the most part as our product is designed for a particular person with specific needs. We would avoid issues for the most part as our product is designed for a particular person with specific needs.

Industrial Design: "Wearable Belt with Adjustable Extension Mechanism for Holding Objects" – Canadian Industrial Design Application No. 184791

Link: [Search Canadian Intellectual Property Office (CIPO) Database]

Summary: This Canadian industrial design registration protects the appearance of a belt-like device with extendable parts that can hold items securely. The design is specifically structured to be ergonomic and adjustable, making it relevant to my project's concept of a belt device with extendable clamps. Although it doesn't protect the functional aspects, the visual appearance and arrangement could influence how we design my device.

Legal Constraints: Since this registration protects the appearance, we need to ensure our device's design differs substantially from this registered design. Altering the overall look, and arrangement of our device could help avoid any legal issues and make our product visually distinct. Customizing the appearance while maintaining functionality will allow our device to respect IP boundaries. We would avoid issues for the most part as our product is designed for a particular person with specific needs.

Project Plan

.	PD E: Project progress presentatio	25 days	2024 October 9 8:30 AM	2024 November 3 11:59 PM	PD D, E.1, E.2	100	NA	25d
PD E: Project progress presentation	PD E.1: Prototype 1	25 days	2024 October 9 8:30 AM	2024 November 3 11:59 PM		100	NA	
PD E: Project progress presentation	PD E.1: Testing	25 days	2024 October 9 8:30 AM	2024 November 3 11:59 PM		100	NA	
PD E: Project progress presentation	PD E.2: Presentation	0 days	2024 October 22 8:30 AM	2024 October 22 11:20 AM	PD E.1	100	NA	
PD E: Project progress presentation	PD E quality check	25 days	2024 October 9 8:00 AM	2024 November 3 11:59 PM		100	NA	
PD E: Project progress presentation	PD E projet plan update	25 days	2024 October 9 8:00 AM	2024 November 3 11:59 PM		100	NA	
PD E: Project progress presentation	PD E submission	13 days	2024 October 22 11:20 AM	2024 November 3 11:59 PM		100	NA	
.	PD F: Design constraints	20 days	2024 October 22 11:20 AM	2024 November 10 11:59 PM	PD E	100	NA	20d
PD F: Design constraints	PD F.1: Design constraints	20 days	2024 October 22 11:20 AM	2024 November 10 11:59 PM		100	NA	
PD F: Design constraints	PD F.2: Prototype 2	20 days	2024 October 22 11:20 AM	2024 November 10 11:59 PM		100	NA	
PD F: Design constraints	PD F.2: Testing	20 days	2024 October 22 11:20 AM	2024 November 10 11:59 PM		100	NA	
PD F: Design constraints	PD F quality check	20 days	2024 October 22 11:20 AM	2024 November 10 11:59 PM		100	NA	
PD F: Design constraints	PD F projet plan update	20 days	2024 October 22 11:20 AM	2024 November 10 11:59 PM		100	NA	
.	Client meet 3	0 days	2024 October 29 8:30 AM	2024 October 29 11:30 AM		100	NA	
.	PD G: Economics and IP considera	7 days	2024 November 11 8:00 AM	2024 November 17 11:59 PM	PD F	100	NA	7d
PD G: Economic and IP considerations	PD G.1: Economics report	7 days	2024 November 11 8:00 AM	2024 November 17 11:59 PM		100	NA	
PD G: Economic and IP considerations	PD G.2: IP report	7 days	2024 November 11 8:00 AM	2024 November 17 11:59 PM		100	NA	
PD G: Economic and IP considerations	PD G quality check	7 days	2024 November 11 8:00 AM	2024 November 17 11:59 PM		100	NA	
PD G: Economic and IP considerations	PD G projet plan update	7 days	2024 November 11 8:00 AM	2024 November 17 11:59 PM		100	NA	
.	PD H: Design day	0 days	2024 November 18 8:00 AM	2024 November 28 11:59 PM	PD G	0	NA	1d
PD H: Design day	Final prototype	11 days	2024 November 18 8:00 AM	2024 November 28 11:59 PM		0	109.3	
PD H: Design day	Design day presentation	11 days	2024 November 18 8:00 AM	2024 November 28 11:59 PM		0	NA	
PD H: Design day	PD H quality check	11 days	2024 November 18 8:00 AM	2024 November 28 3:00 PM		0	NA	
.	Design day	0 days	2024 November 28 10:00 AM	2024 November 28 3:00 PM		0	NA	
.	PD I: User manual	5 days	2024 November 18 8:00 AM	2024 December 3 11:59 PM	PD H	0	NA	5d
PD I: User manual	PD I quality check	5 days	2024 November 18 8:00 AM	2024 December 3 11:59 PM		0	NA	
.	PD J: Final presentation	8 days	2024 November 18 8:00 AM	2024 December 2 5:20 PM	PD I	0	NA	8d
PD J: Final presentation	PD J quality check	8 days	2024 November 18 8:00 AM	2024 December 2 5:20 PM		0	NA	